A Finite-Cutoff Hilbert-Space Model

for Prime–Zero Energy Structure

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Abstract

We construct a finite-cutoff Hilbert-space model for the spectral structure associated with primes and nontrivial zeros of the Riemann zeta function. Given a prime cutoff κ and truncated zero ordinates $\{\gamma_1, \ldots, \gamma_N\}$, we define two finite-dimensional real Hilbert spaces H_{str} (over primes) and H_{null} (over zeros), connected by a linear map $\Phi\colon H_{\text{str}} \to H_{\text{null}}$, and the self-adjoint loop operator $T = \Phi^* \circ \Phi$.

What is proved. We establish the algebraic identity $\Delta := E_{\text{str}} - E_{\text{spec}} = \sum_p c_p^2 \|a_p\|^2 - \|\sum_p c_p a_p\|^2$ by direct expansion.

What is numerical. $\Delta > 0$ and $\eta_{\text{orig}} = \Delta/E_{\text{str}} \in [0.650,\, 0.700]$ for benchmark parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $\sigma \in [0.1, 0.95]$.

What remains open. An analytic proof of $\eta_{\text{orig}} > 0$ remains open.

Structure. The formal core in §§ 2–4 is non-circular and proof-complete; results from Papers 1–2 appear only as context. No use of the Riemann Hypothesis, the GUE conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–4. The model suggests a picture in which $\sigma = \frac{1}{2}$ plays the role of an energy reference point.

Notation and Conventions

Critical line. Throughout this series, σ denotes $\Re(s)$ and the critical line means $\sigma = \frac{1}{2}$. The functional equation symmetry $\xi(s) = \xi(1-s)$ makes $\sigma = \frac{1}{2}$ the natural reference point.

Global parameters. In this paper the reference benchmark values are $\kappa = 53$ (16 active primes $p \le \kappa$), $\varepsilon = 0.05$, and $N = 100$. The formal definitions allow arbitrary fixed $\kappa > 1$, $\varepsilon > 0$, and $N \ge 1$; numerical sections state explicitly when these parameters are varied.

Main objects. The prime space $H_{\text{str}} = \ell^2(S(\kappa))$ and zero space $H_{\text{null}} = \ell^2(\{\gamma_1, \ldots, \gamma_N\})$ are connected by $\Phi\colon H_{\text{str}} \to H_{\text{null}}$, $\Phi(e_p) = a_p$, where $(a_p)_k = e^{-\varepsilon^2 \gamma_k^2/2} \sin(\gamma_k \log p)$. The loop operator is $T = \Phi^* \circ \Phi$[1] and $\eta_{\text{orig}} = 1 - E_{\text{spec}}/E_{\text{str}}$ measures the energy gap.

[1] We call $T = \Phi^*\Phi$ the *loop operator*, following the notation of this series. It is the Gram matrix of the phasor family $\{a_p\}_{p \le \kappa}$ in H_{null}, and should not be confused with loop operators in lattice gauge theory or condensed-matter physics.

Phasor convention. Throughout this paper the prime phasor is $\sin(\gamma_k \log p)$ at $\sigma = \frac{1}{2}$; no additional σ-factor enters the construction. This convention is specific to Papers 1–3 of this series.

Weights. The local prime curvature at the critical line is

$$V_p(\tfrac{1}{2}) \;=\; \frac{4(\log p)^2\, p}{(p-1)^2}.$$

For reference, the renormalized weight of Paper 2 is $f_p := V_p(\frac{1}{2}) - 4(\log p)^2/p = 4(\log p)^2(2p - 1)/(p(p-1)^2)$. The canonical weight in the η-framework of this paper is

$$c_p^\eta \;:=\; \sqrt{V_p(\tfrac{1}{2})}\,.$$

Not to be confused with. $T = \Phi^*\Phi$ (loop operator on H_{str}, this paper) is distinct from $\tilde{T} = \Phi\Phi^*$ (the adjoint loop operator on H_{null}). $D \approx 9.470$ (renormalized energy sum, Paper 2) is distinct from E_{str} (structural energy at benchmark parameters, this paper). The canonical weight $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$ in the η-framework of this paper is *distinct* from Paper 2's renormalized weight $c_p^{\mathrm{ren}} = \sqrt{f_p}$: both in formula ($V_p(\frac{1}{2}) \neq f_p$) and in role ($c_p^\eta$ enters E_{str} inside the η-formula, whereas c_p^{ren} enters $D = \sum_p (c_p^{\mathrm{ren}})^2$ in the Weil renormalization).

Analytical foundations. The construction uses real finite-dimensional inner-product space theory (Gram matrices, positive semi-definite operators) and the prime number theorem via Mertens' formula for weight asymptotics. The explicit formula and Weil's quadratic form [1] provide the infinite-dimensional framework from which the finite-cutoff model is abstracted.

Numerical computations. All numerical results were computed with `mpmath` at 50 decimal digits of working precision; Riemann zero ordinates γ_k were evaluated via `mpmath.zetazero` and `numpy` was used for matrix operations. Verification code is available at https://github.com/utehrani/analysislab-nt.

1 Introduction

1.1 Background and Motivation

The present paper is the third in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function $\zeta(s)$. The formal core in §§ 2–4 is non-circular and proof-complete.

Paper 1 [2] introduced the curvature function $H_\xi(\sigma, t) := \frac{d^2}{d\sigma^2} \log |\xi(\sigma + it)|^2$ and established the divergence $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 2(\log \kappa)^2 \to \infty$ at $\sigma = \frac{1}{2}$, contrasted with convergence for $\sigma > \frac{1}{2}$.

Paper 2 [3] connected this local curvature to the Weil explicit functional, constructing admissible test functions and establishing the convergent weight sum $D = \sum_p (c_p^{\text{ren}})^2 \approx 9.470$. These results appear only as motivation in the introduction and in the contextual remark of § 2; the formal core does not depend on them.

The present paper makes the geometric structure underlying both papers explicit: two finite-dimensional Hilbert spaces H_{str} and H_{null}, connected by a linear map Φ, and a self-adjoint loop operator $T = \Phi^* \circ \Phi$.

Several established programs seek a spectral interpretation of the Riemann zeros. Berry and Keating [4] studied a semiclassical analogue via quantum chaos; Connes [5] realizes the zeros through a trace formula in noncommutative geometry; de Branges [6] approached the problem through Hilbert spaces of entire functions; Burnol [7] constructs a conductor operator on Sonin spaces directly from the explicit formula. These programs work in spectral or infinite-dimensional analytic frameworks. The present model is structurally distinct: H_{str} and H_{null} are finite-dimensional and real, no spectral postulate is made, and the γ_k enter as labels of basis vectors, not as eigenvalues. For a taxonomy of physics-of-RH approaches see Schumayer–Hutchinson [8].

The Gram-matrix construction $T_{pq} = \langle a_p, a_q \rangle$ from damped sine vectors appears to be new; conceptually it may be viewed as a discrete analogue of spectral interpretations of the explicit formula, as studied by Goldfeld [9]. The test functions of Paper 2 connect to adelic Schwartz functions as in Tate's thesis [10]; the Weil positivity criterion [1, 11] provides the outer framework. Bombieri–Lagarias [12] establish that finite-truncation positivity arguments are not zeta-specific; the arithmetic content of this paper lies entirely in the numerical value of $\eta_{\text{orig}} > 0$, not in the algebraic identity $\Delta \geq 0$. Bender–Brody–Müller [13] (with rebuttal [14]) propose a non-Hermitian Hamiltonian whose reality of spectrum would imply $\sigma = \frac{1}{2}$; our approach treats $\sigma = \frac{1}{2}$ as a finite-cutoff variance reference point, motivated by functional-equation symmetry and the local-curvature divergence from **Paper 1** [2], rather than spectral assumptions. Ionescu [15] forms a related prime-character pairing without the Gaussian regulator or the bipartite operator framing. Bombieri [16] gives a systematic analysis of the Weil quadratic functional in the context of obstructions to RH; our finite-cutoff model implements a truncated version of this functional without addressing the infinite-dimensional obstructions.

The renormalized framework of Paper 2 and the η-framework of this paper use different normalizations; a precise identification remains open (Open Problem 5.5).

1.2 Main Results

We define two finite-dimensional real Hilbert spaces H_{str} (over primes $p \le \kappa$) and H_{null} (over zero ordinates $\gamma_1, \ldots, \gamma_N$), connected by the linear map $\Phi : H_{\mathrm{str}} \to H_{\mathrm{null}}$ built from Gaussian-damped prime resonance vectors (§ 3). The self-adjoint loop operator $T = \Phi^* \circ \Phi$ is the Gram matrix of these vectors (§ 3).

We prove the algebraic identity $\Delta := E_{\mathrm{str}} - E_{\mathrm{spec}}$ by direct expansion (Theorem 4.3). Numerically, $\eta_{\mathrm{orig}} = \Delta / E_{\mathrm{str}} \in [0.650, 0.700]$ for benchmark parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $\sigma \in [0.1, 0.95]$ (Observation 4.4). An analytic proof of $\eta_{\mathrm{orig}} > 0$ remains open (Open Problem 5.1).

1.3 Reader Contract

This paper *proves*, by direct algebraic expansion, the variance identity $\Delta = E_{\mathrm{str}} - E_{\mathrm{spec}}$ (Theorem 4.3) and the basic properties of the loop operator $T = \Phi^* \circ \Phi$: self-adjointness, positive semi-definiteness, and σ-invariance of the normalized Gram matrix (Proposition 3.5). These are unconditional.

This paper *establishes numerically* that $\eta_{\mathrm{orig}} > 0$ for all tested parameter combinations, with $\eta_{\mathrm{orig}} \in [0.650, 0.700]$ at benchmark parameters (§ 4). The finite-grid spectral bound $B^{E}_{\max} < 1$ and $B^{\|c\|}_{\max} < 1$ and the analytic explanation of the value of $B^{\|c\|}_{\max}$ (Problem 5.4) are numerical observations and open problems respectively.

This paper *leaves open* the analytic proof of $\eta_{\mathrm{orig}} > 0$ (Problem 5.1), the spectral bound for canonical weights (Problem 5.2), the analytical cancellation bound (Problem 5.3), the analytic explanation of $B^{\|c\|}_{\max}$ (Problem 5.4), and the renormalized shell energy (Problem 5.5).

2 Two Hilbert Spaces

We work throughout with real Hilbert spaces; all vectors and operators have real entries. Complex phasor language appears in § 3.1 as heuristic motivation only; all formal objects $(H_{\mathrm{str}}, H_{\mathrm{null}}, \Phi, T)$ are real.

Throughout this paper, $\kappa > 1$ is a fixed real cutoff parameter, $S(\kappa) := \{p \text{ prime} : p \le \kappa\}$ the set of active primes, and $\{\gamma_k\}_{k=1}^{N}$ the first N positive imaginary parts of nontrivial zeros of $\zeta(s)$, ordered $0 < \gamma_1 \le \gamma_2 \le \cdots$. The Gaussian damping parameter $\varepsilon > 0$ is fixed throughout.

Definition 2.1 (Prime space H_{str}).

$$H_{\mathrm{str}} := \ell^2(S(\kappa))$$

with basis $\{e_p : p \in S(\kappa)\}$, *standard inner product* $\langle a, b\rangle_{\mathrm{str}} = \sum_p a_p b_p$, *and* $\dim H_{\mathrm{str}} = |S(\kappa)| = \pi(\kappa)$. *The* structural energy *of a weight vector* $c \in H_{\mathrm{str}}$ *is* $E_{\mathrm{str}}(c) = \sum_p c_p^2 \|a_p\|^2$ *(defined in §4 after* Φ *is introduced).*

Definition 2.2 (Zero space H_{null}).

$$H_{\text{null}} := \ell^2(\{\gamma_1, \ldots, \gamma_N\})$$

with basis $\{f_k : k = 1, \ldots, N\}$*, inner product* $\langle u, v\rangle_{\text{null}} = \sum_{k=1}^{N} u_k v_k$ *(unweighted), and* $\dim H_{\text{null}} = N$*.*

Remark (Why $\sigma = \frac{1}{2}$ is a distinguished reference point)**.** *This remark provides context; it is not used in §§ 3–5.*

Two independent proven reasons support this choice:

(i) Symmetry *[proven]:* $\sigma = \frac{1}{2}$ *is the unique fixed point of* $s \mapsto 1 - s$*, the symmetry of the completed zeta function* $\xi(s) = \xi(1 - s)$*.*

(ii) Curvature singularity *[proven, Papers 1–2]:* $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 2(\log \kappa)^2 \to \infty$ *as* $\kappa \to \infty$*, while* $H_{\text{local}}(\sigma, \kappa) \to C(\sigma) < \infty$ *for all* $\sigma > \frac{1}{2}$*.*

$\sigma = \frac{1}{2}$ *is thus a distinguished reference point by symmetry and curvature divergence. Energy near-equality (*$E_{\text{str}} \approx E_{\text{spec}}$*) is NOT observed in the finite-cutoff model (see §5).*

3 The Linear Map Φ and the Loop Operator T

Convention. Gaussian damping is carried by the vectors: $a_p = (e^{-\varepsilon^2\gamma_k^2/2}\sin(\gamma_k \log p))_k$. The H_{null} inner product is unweighted: $\langle u, v\rangle = \sum_k u_k v_k$.

3.1 Motivation for the resonance vectors

The formal definition is Definition 3.2 below; this subsection provides intuition.

The prime resonance vector at zero ordinate γ_k and prime p arises as the imaginary part of the prime phasor $\varphi_p(u) = e^{iu\log p}$ evaluated at $u = \gamma_k$, with Gaussian regularization for convergence:

$$(a_p)_k = e^{-\varepsilon^2\gamma_k^2/2} \cdot \sin(\gamma_k \log p).$$

The k-th component measures how strongly prime p resonates with zero ordinate γ_k.

Definition 3.1 (Prime resonance vectors)**.** *For each prime* $p \in S(\kappa)$*, the* canonical resonance vector *at* $\sigma = \frac{1}{2}$ *is*

$$a_p^{\text{can}} := a_p := \left(e^{-\varepsilon^2\gamma_k^2/2}\sin(\gamma_k \log p)\right)_{k=1}^{N} \in H_{\text{null}}.$$

At general σ*, the resonance vectors depend on* σ *via*

$$a_p^{(\sigma)} := p^{-(\sigma-1/2)} \cdot a_p^{\text{can}}.$$

The vectors $a_p = a_p^{\text{can}} = a_p^{(1/2)}$ *at* $\sigma = \frac{1}{2}$ *are the canonical representatives used in all primary definitions.*

Definition 3.2 (Linear map Φ).

$$\Phi\colon H_{\mathrm{str}} \to H_{\mathrm{null}}, \qquad \Phi(e_p) := a_p^{\mathrm{can}}.$$

Extended linearly: $\Phi(c) = \sum_p c_p a_p^{\mathrm{can}}$ *for any* $c = \sum_p c_p e_p \in H_{\mathrm{str}}$.

We define Φ as a linear map; no injectivity claim is made for the finite-cutoff model.

Definition 3.3 (Gram matrices). *The* unnormalized Gram matrix *(canonical) is*

$$G_{pq}^{\mathrm{un}} := G_{pq}^{\mathrm{un,can}} := \langle a_p^{\mathrm{can}}, a_q^{\mathrm{can}} \rangle_{H_{\mathrm{null}}} = \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2} \sin(\gamma_k \log p) \sin(\gamma_k \log q).$$

The σ-dependent extension is $G^{\mathrm{un}}(\sigma)_{pq} = p^{-(\sigma-1/2)} q^{-(\sigma-1/2)} G_{pq}^{\mathrm{un}}$. *The* normalized Gram matrix *is*

$$G_{pq}^{\mathrm{norm}} := \frac{G_{pq}^{\mathrm{un}}}{\|a_p^{\mathrm{can}}\| \cdot \|a_q^{\mathrm{can}}\|} = \left\langle \frac{a_p^{\mathrm{can}}}{\|a_p^{\mathrm{can}}\|}, \frac{a_q^{\mathrm{can}}}{\|a_q^{\mathrm{can}}\|} \right\rangle_{H_{\mathrm{null}}}.$$

The matrix G^{norm} is σ-invariant (Proposition 3.5(iv)).

Definition 3.4 (Loop operator T).

$$T := \Phi^* \circ \Phi\colon H_{\mathrm{str}} \longrightarrow H_{\mathrm{str}}.$$

The matrix representation of T in the prime basis is $T_{pq} = G_{pq}^{\mathrm{un}}$. T is thus identical to the canonical Gram matrix G^{un}, and is explicitly constructed from prime resonance vectors.

Proposition 3.5 (Properties of T; proved). (i) Self-adjoint: $T^* = (\Phi^*\Phi)^* = \Phi^*\Phi = T$.

(ii) Positive semi-definite: $\langle Tc, c \rangle = \|\Phi c\|^2 \geq 0$ *for all* $c \in H_{\mathrm{str}}$.

(iii) Non-negative eigenvalues: *All eigenvalues* $\lambda_j \geq 0$.

(iv) σ-invariance of G^{norm} *[by direct calculation]: For the σ-dependent family $a_p^{(\sigma)}$, the normalized Gram matrix G_{pq}^{norm} is independent of σ.* Proof. *The scalar factor $p^{-(\sigma-1/2)}$ cancels in the normalized ratio.* □

Remark (Spectral structure of T). *The eigenvalues $\lambda_j \geq 0$ of $T = \Phi^* \circ \Phi$ are the squares of the singular values $\sigma_j(\Phi)$: $\lambda_j = \sigma_j(\Phi)^2$. The eigenvalues λ_j are built from the zero ordinates γ_k appearing in the definition of a_p^{can}, but are* not *directly the zeros γ_k themselves. This distinguishes T from the classical Hilbert-Pólya ansatz: T is explicitly constructed, not postulated.*

4 The Variance Identity and η_{orig}

Convention. Objects without σ-argument refer to the canonical ($\sigma = \frac{1}{2}$) version throughout § 4.

Definition 4.1 (Canonical and σ-dependent energies). *For a weight vector $c \in H_{\mathrm{str}}$ with components c_p:*

Canonical quantities *(at $\sigma = \frac{1}{2}$):*

$$D^{\mathrm{can}} := \operatorname{diag}\big(\|a_p^{\mathrm{can}}\|^2\big)_{p\in S(\kappa)}, \qquad E_{\mathrm{str}}^{\mathrm{can}} := c^\top D^{\mathrm{can}} c = \sum_p c_p^2 \|a_p^{\mathrm{can}}\|^2,$$

$$E_{\mathrm{spec}}^{\mathrm{can}} := \|\Phi c\|^2 = c^\top G^{\mathrm{un}} c.$$

σ-dependent extensions:

$$D(\sigma)_{pp} := p^{-2(\sigma-1/2)} \cdot D_{pp}^{\mathrm{can}}, \qquad E_{\mathrm{str}}(\sigma) := \sum_p c_p^2\, p^{-2(\sigma-1/2)} \|a_p^{\mathrm{can}}\|^2,$$

$$E_{\mathrm{spec}}(\sigma) := c^\top G^{\mathrm{un}}(\sigma)\, c.$$

The canonical weight vector *in the η-context is*

$$c_p^\eta \;:=\; \sqrt{V_p(\tfrac{1}{2})} \;=\; \sqrt{\frac{4(\log p)^2\, p}{(p-1)^2}}, \qquad p \in S(\kappa).$$

Remark (Canonical weights — normative form)**.** *The canonical weight $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$ is the normative form in the η-framework. For large p, $c_p^\eta \approx 2(\log p)/\sqrt{p}$; this differs from the Weil weight $c_p^{\mathrm{ren}} = \sqrt{f_p}$ both in formula and asymptotic behaviour.*

Assumption 4.2 (Positive definiteness of D)**.** *Throughout this paper we assume $D_{pp}^{\mathrm{can}} = \|a_p\|^2 > 0$ for all $p \in S(\kappa)$. This holds in all numerical experiments conducted here.*

Theorem 4.3 (Algebraic identity; proved)**.** *The following equality holds by direct expansion:*

$$\Delta := E_{\mathrm{str}} - E_{\mathrm{spec}} = \sum_p c_p^2 \|a_p\|^2 - \Big\|\sum_p c_p a_p\Big\|^2.$$

Proof. Expand $E_{\mathrm{spec}} = \|\Phi c\|^2 = \big\langle \sum_p c_p a_p, \sum_q c_q a_q \big\rangle = \sum_{p,q} c_p c_q \langle a_p, a_q\rangle = c^\top G^{\mathrm{un}} c$. Then $\Delta = c^\top D^{\mathrm{can}} c - c^\top G^{\mathrm{un}} c$, which is the stated identity. □

Remark (Algebraic identity vs. arithmetic content)**.** *The inequality $\Delta \geq 0$ is* not *an algebraic consequence of Theorem 4.3: $T = \Phi^*\Phi$ is positive semi-definite, but $\Delta = c^\top(D^{\mathrm{can}} - G^{\mathrm{un}})c$ has indefinite sign for a general weight vector c. The non-trivial arithmetic content is the numerical lower bound $\eta_{\mathrm{orig}} \geq 0.650 > 0$ for the canonical weight $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$ at the reference parameters, established in the observations of this section.*

Observation 4.4 (Non-negativity, numerical)**.**

$$\Delta \geq 0 \quad \textit{equivalently} \quad \eta_{\mathrm{orig}} \geq 0$$

holds for all tested parameters with canonical weights $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$ (Definition 4.1). [Numerical only — no unconditional proof for general c.]

Note: For all *weight vectors c, the inequality $\eta_{\mathrm{orig}}(c) \geq 0$ would be equivalent to $\lambda_{\max}\big((D^{\mathrm{can}})^{-1/2} T (D^{\mathrm{can}})^{-1/2}\big) \leq 1$. This universal bound is false on the tested cutoffs: on those cutoffs, $\lambda_{\max}(T_{\mathrm{ren}})/\pi(\kappa)$ is nu-*

merically of order 0.39*, so* $\lambda_{\max}(T_{\text{ren}}) > 1$ *already at finite tested cutoffs. The relevant open problem is therefore not a universal bound in* c*, but the canonical-weight Rayleigh quotient condition in Open Problem 5.2.*

Cauchy-Schwarz does **not** guarantee $\eta_{\text{orig}} \geq 0$*: the correct identity is* $\Delta = c^\top(D^{\text{can}} - G^{\text{un}})c$*; the sign of* Δ *is not determined by the algebraic identity alone.*

Definition 4.5 (η_{orig})**.**

$$\eta_{\text{orig}} := 1 - \frac{E_{\text{spec}}}{E_{\text{str}}} = 1 - \frac{\|\Phi c\|^2}{E_{\text{str}}}.$$

Boundary cases: $\eta_{\text{orig}} = 0$ *iff* $E_{\text{spec}} = E_{\text{str}}$*;* $\eta_{\text{orig}} = 1$ *iff* $\Phi c = 0$*.*

Remark (σ-dependence via rescaled weights)**.** *The entire* σ*-dependence of* $\eta_{\text{orig}}(\sigma)$ *is carried by the rescaled weight vector* $\tilde{c}_p(\sigma) := c_p \cdot p^{-(\sigma-1/2)}$*:*

$$\eta_{\text{orig}}(\sigma) = 1 - \frac{\tilde{c}(\sigma)^\top G^{\text{un}}\, \tilde{c}(\sigma)}{\tilde{c}(\sigma)^\top D^{\text{can}}\, \tilde{c}(\sigma)}.$$

Proof. *Since* $G^{\text{un}}(\sigma)_{pq} = p^{-(\sigma-1/2)} q^{-(\sigma-1/2)} G^{\text{un}}_{pq}$*, one has* $c^\top G^{\text{un}}(\sigma) c = \tilde{c}^\top G^{\text{un}} \tilde{c}$ *and* $E_{\text{str}}(\sigma) = \tilde{c}^\top D^{\text{can}} \tilde{c}$*.* ☐

Numerical evidence. All computations use $N = 100$ zero ordinates at the reference parameters; the algebraic identity (Theorem 4.3) is verified to rounding error $< 10^{-14}$. Verification scripts are available in the companion repository [17].

Observation 4.6 (η_{orig} positive across all tested parameters, numerical)**.** $\eta_{\text{orig}} > 0$ *in all tested parameter combinations. For the benchmark slice* $(\kappa, \varepsilon, N) = (53, 0.05, 100)$*,* $\sigma \in [0.1, 0.95]$*:*

$$\eta_{\text{orig}} \in [0.650,\ 0.700].$$

Note: the range $[0.650, 0.700]$ *is strongly* ε*-dependent; no universal range over all* (κ, ε) *is claimed.*

The positivity check was performed for $\kappa \in \{23, 53, 101, 199\}$, $\varepsilon = 0.05$, $N = 100$, and the σ-grid $\{0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 0.95\}$. The table below displays the benchmark slice $(\kappa, \varepsilon, N) = (53, 0.05, 100)$: on this grid $\eta_{\text{orig}}(\sigma)$ has a shallow local maximum near $\sigma \approx 0.20$ and then decreases. No monotonicity theorem is claimed.

σ	η_{orig}
0.10	0.69928647
0.20	0.69989546
0.30	0.69866521
0.40	0.69558817
0.50	0.69078176
0.60	0.68441702
0.70	0.67662272
0.80	0.66742445
0.90	0.65675975
0.95	0.65085191

Observation 4.7 (Finite-grid mode diagnostics $B^E_{\max}$ and $B^{\|c\|}_{\max} < 1$, numerical)**.** *The following computations use $\kappa \in \{23, 53, 101, 199\}$, $\varepsilon \in \{0.02, 0.05, 0.10\}$, $\sigma \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$, $N = 100$.*

Set $\tilde{c}(\sigma)_p := c_p \cdot p^{-(\sigma-1/2)}$ (rescaled weight vector). Two mode-contribution diagnostics are computed, where λ_j, u_j denote eigenvalues and eigenvectors of $T = G^{\mathrm{un}}$ (at $\sigma = \frac{1}{2}$, canonical):

$$B^E_j(\sigma) := \frac{\lambda_j\, |\langle \tilde{c}(\sigma), u_j\rangle|^2}{\tilde{c}(\sigma)^\top D^{\mathrm{can}} \tilde{c}(\sigma)}, \qquad B^{\|c\|}_j := \frac{\lambda_j\, |\langle c, u_j\rangle|^2}{\|c\|^2}.$$

If no eigenvalue exceeds 1, we set $B_{\max} := 0$ by convention. The E_{str}-normalized diagnostic satisfies $\sum_j B^E_j(\sigma) = 1 - \eta_{\mathrm{orig}}(\sigma)$ [algebraic, all σ]; the $\|c\|^2$-normalized diagnostic does not. Numerically at $(\kappa, \varepsilon) = (53, 0.05)$:

$$B^E_{\max}(\sigma) := \max_{j:\lambda_j>1} B^E_j(\sigma) \in [0.115,\ 0.127] \textit{ for } \sigma \in [0.3, 0.7], \quad < 1$$
$$B^{\|c\|}_{\max} := \max_{j:\lambda_j>1} B^{\|c\|}_j \approx 0.0735, \quad < 1$$

Both satisfy their respective maximum < 1 for all tested parameters.

Selected values of $B^{\|c\|}_{\max}$ (the tabulated diagnostic):

κ	$\varepsilon = 0.02$	$\varepsilon = 0.05$	$\varepsilon = 0.10$
23	0.2975	0.0112	0.0000
53	0.3813	0.0735	0.0000
101	0.1241	0.0483	0.0000
199	0.1546	0.0483	0.0107

Observation 4.8 ($B^{\|c\|}_{\max}$ σ-invariant by construction, finite-grid numerical)**.** *The diagnostic $B^{\|c\|}_{\max} \approx 0.0735$ for $(\kappa, \varepsilon) = (53, 0.05)$ is independent of σ by construction: both c and the eigenvectors u_j of T are fixed at $\sigma = \frac{1}{2}$. The analytical explanation of the value 0.0735 remains open (Open Problem 5.4).*

Remark (Cancellation mechanism, numerical analysis). *The mechanism underlying* $B_{\max} < 1$ *is a cancellation between the monotone*

weight vector $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$ *and the oscillatory eigenvectors* u_j *of* G^{un}*: for* $\kappa = 53$*, eigenvectors* $u_1, \ldots, u_4$ *each exhibit 10 sign changes along the prime ordering. Cancellation rates 67–97% observed for* $\kappa \in \{23, 53, 101, 199\}$*. Formal proof: Open Problem 5.3.*

5 Open Problems

Unless otherwise stated, the open problems below concern the finite-cutoff family with $T = T(\kappa, \varepsilon, N)$, $D^{\mathrm{can}} = D^{\mathrm{can}}(\kappa, \varepsilon, N)$, and canonical weights $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$. Numerical evidence uses $N = 100$ and the explicitly stated grids in (κ, ε). A uniform theorem in κ, ε, or an asymptotic regime is a stronger statement and is not claimed unless specified.

Open Problem 5.1 (Analytic positivity). *Prove analytically that* $\eta_{\mathrm{orig}} > 0$ *for the canonical weight vector* $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$*. Numerically confirmed:* $\eta_{\mathrm{orig}} \in [0.650, 0.700]$ *for* $(\kappa, \varepsilon, N) = (53, 0.05, 100)$*,* $\sigma \in [0.1, 0.95]$ *(§ 4).*

Open Problem 5.2 (Spectral bound for canonical weights). *Let*

$$T_{\mathrm{ren}} := (D^{\mathrm{can}})^{-1/2} T (D^{\mathrm{can}})^{-1/2}, \qquad x := (D^{\mathrm{can}})^{1/2} c,$$

where $c = c^\eta$ *is the canonical weight vector. For this fixed canonical weight,* $\eta_{\mathrm{orig}} > 0$ *is equivalent to*

$$\frac{\langle T_{\mathrm{ren}} x, x\rangle}{\|x\|^2} < 1.$$

The universal bound $\lambda_{\max}(T_{\mathrm{ren}}) < 1$*, which would imply positivity for every weight vector, is not true on the tested cutoffs and is not claimed here. Prove the above Rayleigh quotient condition analytically for* $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$*. For all tested* $\kappa \in \{23, 53, 101, 199, 503, 1009\}$ *the condition holds numerically. The vector* x *appears nearly orthogonal to the dominant eigenmode of* T_{ren} *(localized at* $\gamma_1 = 14.13\ldots$*); proving this near-orthogonality is the core open problem.*

Open Problem 5.3 (Analytical cancellation bound). *Prove analytically that*

$$B_{\max}^{\|c\|} := \max_{j:\,\lambda_j > 1} \lambda_j \frac{|\langle c, u_j\rangle|^2}{\|c\|^2} < 1,$$

or equivalently, for all j *with* $\lambda_j > 1$*:*

$$\frac{|\langle c, u_j\rangle|^2}{\|c\|^2} \le \frac{C_B}{\lambda_j}, \quad C_B < 1.$$

Note: since the tested data show growth of $\lambda_{\max}(T_{\mathrm{ren}})$ *with* $\pi(\kappa)$*, a uniform projection bound* $|\langle c, u_j\rangle|^2/\|c\|^2 \le C_1 \ll 1$ *does not in itself imply* $B_{\max}^{\|c\|} < 1$*; the growth of* λ_j *must also be absorbed. Approach: Abel summation for* $c_p^\eta = \sqrt{V_p(\frac{1}{2})} \sim 2(\log p)/\sqrt{p}$ *against oscillatory partial sums* $\sum_{q \le p} (u_j)_q$*, without reference to RH. Numerically:* $B_{\max}^{\|c\|} < 0.382$ *for the tested grid,* $\kappa \in \{23, 53, 101, 199\}$*,* $\varepsilon \in \{0.02, 0.05, 0.10\}$*,* $N = 100$*.*

Open Problem 5.4 (Analytic explanation of $B_{\max}^{\|c\|}$)**.** *The diagnostic* $B_{\max}^{\|c\|} := \max_{j:\lambda_j>1} \lambda_j |\langle c, u_j\rangle|^2/\|c\|^2$ *uses the fixed canonical weight* $c = c_p^\eta$ *and canonical eigenvectors* u_j *of* T *at* $\sigma = \frac{1}{2}$*. It therefore carries no* σ*-dependence by definition, and its* σ*-invariance is a structural feature of the formula. What remains open is the analytical explanation of the numerical value* $B_{\max}^{\|c\|} \approx 0.0735$ *at* $(\kappa, \varepsilon) = (53, 0.05)$*.*

Open Problem 5.5 (Renormalized shell energy)**.** *In the Paper 2 normalization framework, define the renormalized shell energy using canonical effective coupling* $N_{\text{eff}} = \pi(\kappa)$*:*

$$H_{\text{ren}}(\kappa, \varepsilon(\kappa)) := H_{\text{local}}\big(\tfrac{1}{2}, \kappa\big) - 2(\log\kappa)^2 = O(\log\kappa)$$

(Paper 2, §5), where $\pi(\kappa)$ *is the prime-counting function at cutoff* κ*. Compute* $H_{\text{ren}}(\kappa, \varepsilon(\kappa))$ *explicitly and determine whether it has a stable limiting or asymptotic form. This problem belongs to the Paper 2 normalization and is independent of the algebraic identity proved in Section 4.*

Scope and limitations. The algebraic identity $\Delta = E_{\text{str}} - E_{\text{spec}} = c^\top D^{\text{can}} c - c^\top G^{\text{un}} c$ is a direct consequence of the definitions. It does *not* imply $\Delta \geq 0$: for a general weight vector c, the form $D^{\text{can}} - G^{\text{un}}$ need not be positive semidefinite. The sign statement $\Delta \geq 0$ is established here only numerically for the canonical weight $c_p^\eta = \sqrt{V_p(\frac{1}{2})}$ on the tested parameter grids. Bombieri–Lagarias [12] establish that finite-truncation positivity is generically not zeta-specific; the arithmetic content lies in the specific numerical value $\eta_{\text{orig}} \in [0.650, 0.700]$ and in the five open problems. No claim is made toward the Riemann Hypothesis.

6 Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. At no point is the Riemann Hypothesis assumed. The zero ordinates γ_k enter the construction of Φ as numerically specified inputs, and no property of their distribution requiring the Riemann Hypothesis is assumed or used.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The loop operator $T = \Phi^*\Phi$ is built explicitly from the prime resonance vectors $\{a_p\}_{p\leq\kappa}$. Its construction postulates neither GUE statistics, Montgomery's pair correlation conjecture, nor the Hilbert–Pólya conjecture. The eigenvalues λ_j of T are the squares of the singular values of Φ and are not identified with zero ordinates γ_k.

Status separation maintained. The variance identity $\Delta = E_{\text{str}} - E_{\text{spec}}$ is an algebraic theorem; the sign statement $\Delta \geq 0$ is established numerically on the tested grid. These are distinct levels and are flagged separately throughout. No claim relating η_{orig} to RH is proved or used.

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Call for Feedback

This paper is part of an ongoing open research initiative toward the Riemann Hypothesis. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

Zenodo (https://zenodo.org/search?q=Ulrich+Tehrani) and GitHub (https://github.com/utehrani/analysislab-nt).

Topics of particular interest include: whether $\eta_{\mathrm{orig}} > 0$ is an expected consequence of known equidistribution results for the phases $\{\gamma_k \log p \bmod 2\pi\}$, and whether the observed near-orthogonality of the weight vector to the dominant eigenmode of T connects to standard cancellation phenomena in partial sums involving primes and zeros.

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